

Alexandr Savca

Security Engineer and Systems Programmer.

Selected Achievements

- Five CVE disclosures affecting Ruby libraries and network services
 - Wireshark contributions extending Oracle TNS protocol dissection
 - Linux kernel input subsystem contribution
 - Systems engineering for the [Zeppel-Lin](#) GNU/Linux project
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Selected Research & Writing

The Fieldbook (2026)

Author of [The Fieldbook](#), a technical book on systems failure analysis: semantic authority, contracts and invariants, boundary failure, semantic drift, failure autopsy, intervention, verification, and repair. It treats technical components, operators, and institutional memory as one coupled system when they materially affect how the system fails or remains secure.

Experience

Information Security Engineer (Freelance)

Jan 2020 – Present

- Perform source-code review and application security assessments
 - Conduct penetration testing and vulnerability research
 - Develop exploit code and verification tooling
 - Build and maintain security tooling and automation
 - Perform scanning, patching, and remediation
 - Contribute patches to security tools and Linux-based systems
 - Maintain SOP documentation for operational continuity
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Information Security Engineer — Salt Edge

Jan 2018 – Nov 2019

- Performed white-box and black-box security testing
 - Conducted manual code review and threat modeling
 - Enforced secure design principles across projects
 - Supported remediation of vulnerabilities
 - Designed and operated bug bounty program
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System Programmer (Freelance)

Sep 2016 – Feb 2017

- Reverse engineered network protocols
 - Developed Zeek (Bro) IDS plugins
 - Analyzed Oracle TNS protocol behavior
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System Administrator (Freelance)

2012 – 2017

- Administered Linux-based infrastructure
 - Tuned kernel and system performance
 - Managed DNS, VPN, web, database, and mail services
 - Developed automation scripts (C, shell, Perl, Tcl/Tk)
 - Performed network analysis and troubleshooting
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Education

Master's Degree — Database Administration and Web Technologies

2014 – 2016 Thesis: SQLite replicator for UNIX systems

Master's Degree — Psychological Counseling

2012 – 2014 Thesis: Coping behavior in adolescents under varying family conditions